# Organic Cume
# April 26, 2002

Give an example and a detailed mechanism for three of the following four transition metal mediated reactions. Name each mechanistic step with an appropriate term (reductive elimination, etc). Give the oxidation state of one intermediate (of your choice) in each mechanism.

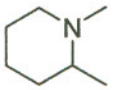
1) Heck Reaction

2) Dötz Reaction

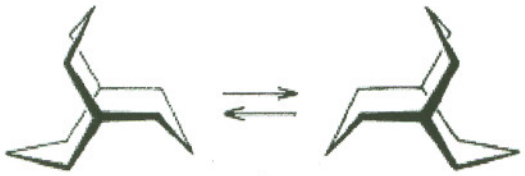
3) Pauson-Kand Reaction

4) Alkyne cyclotrimerization.

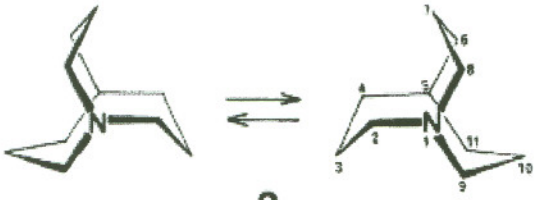
2) The compound depicted below shows peaks in the mass spectrum at $m/z = 98$ and 70. Explain these results by providing plausible ***mechanisms*** for the fragmentation patterns upon electron impact ionization.



3) The molecules manxane (**1**) and manxine (**2**) exhibit some rather unusual chemistry (the names of these molecules comes from the resemblance to the Coat of Arms of the Isle of Man, as shown below). The X-ray crystal structures of both **1** and **2** show a considerable flattening at the bridgehead atoms, i.e. the C-C-C bridgehead angles are 113.5-114.8 ° and the C-N-C bridgehead angles are 115.0-115.9 ° (see *Acc. Chem. Res.* **1979**, page 423). Based on these findings, comment on the following:



**1**



**2**

a) The ionization potential of **2** is at a very low energy, and **1** reacts with oxygen even when stored in a brown screw-capped vial in the absence of light.

b) Would you expect the proton affinity of **2** to be greater or less than that of tri-n-propyl amine?